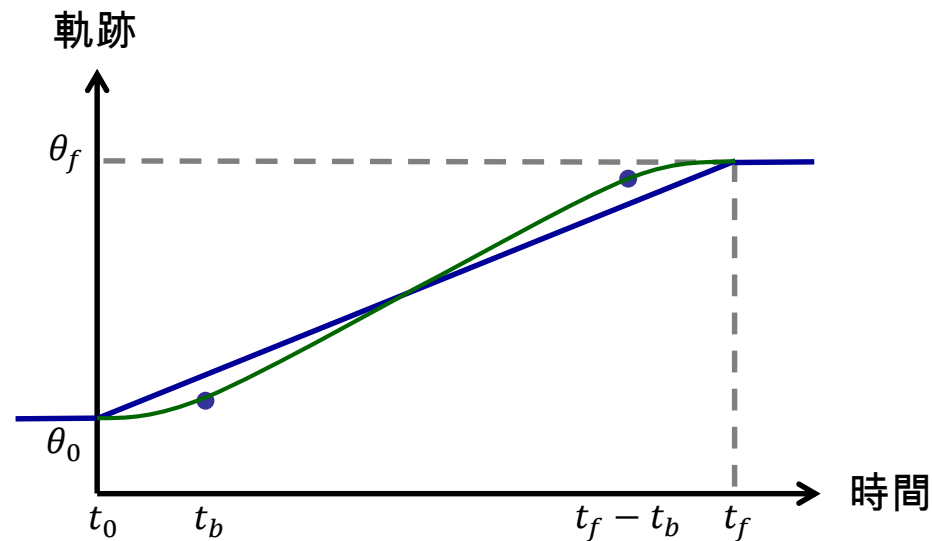


Linear Function with Parabolic Blends說明-1

□ 緣由

- ◆ 在許多類型的任務上均需要使用 直線軌跡
- ◆ 軌跡中若包含多個直線段軌跡，線段間轉折點速度不連續
- ◆ 解決方式：將直線段兩端修正為二次方程式，讓速度軌跡smooth



(assume $\dot{\theta}_0 = \dot{\theta}_f = 0$)

Linear Function with Parabolic Blends 說明-2

□ 規劃方式

◆ linear (直線, 一次多項式)

○ 等速

$$\dot{\theta} = \frac{\theta_h - \theta_b}{t_h - t_b} = \dot{\theta}_{t_b} \quad \dots \textcircled{1}$$

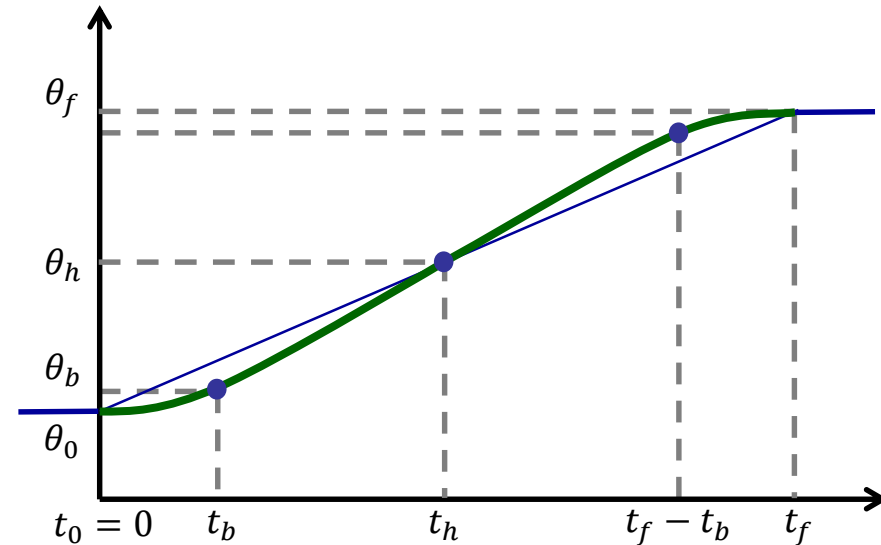
◆ Parabolic (二次多項式)

○ 等加速度

$$\begin{aligned} \theta(t) &= \theta_0 + \dot{\theta}_0 t + \frac{1}{2} \ddot{\theta} t^2 \\ \dot{\theta}(t) &= \dot{\theta}_0 + \ddot{\theta} t \end{aligned}$$

↑
加速度

$$\dot{\theta}(t_b) = \ddot{\theta} t_b \quad \dots \textcircled{2}$$



$$t_h = \frac{1}{2}t_f \quad \theta_h = \frac{\theta_f + \theta_0}{2}$$

$$\text{assume } \dot{\theta}_0 = \dot{\theta}_f = 0$$

Linear Function with Parabolic Blends 說明-3

- ◆ 交界處速度需要連續

$$\textcircled{2} = \textcircled{1}$$

$$\ddot{\theta}t_b = \dot{\theta}_{t_b} = \frac{\theta_h - \theta_b}{t_h - t_b} = \frac{\frac{\theta_f + \theta_0}{2} - (\theta_0 + \frac{1}{2}\ddot{\theta}t_b^2)}{\frac{t_f}{2} - t_b} = \frac{\theta_f - \theta_0 - \ddot{\theta}t_b^2}{t_f - 2t_b}$$

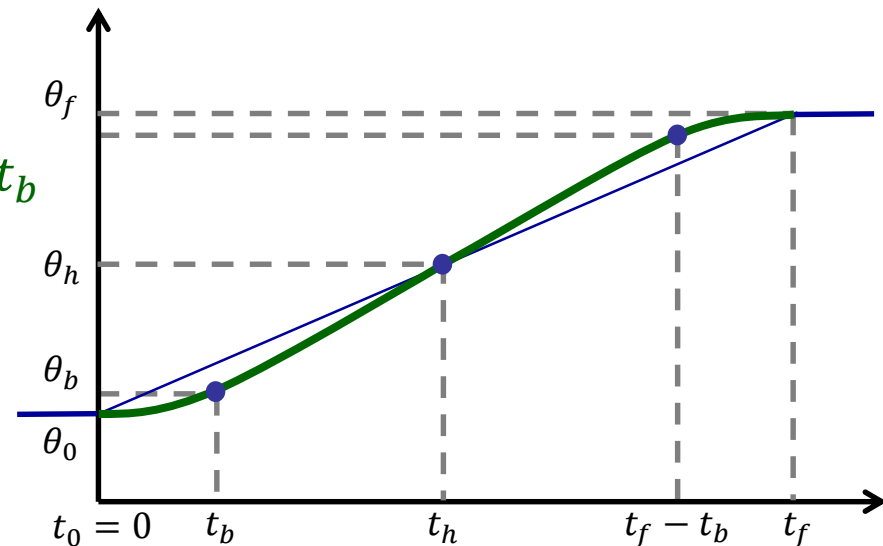
$$\ddot{\theta}t_b^2 - \ddot{\theta}t_ft_b + (\theta_f - \theta_0) = 0$$

$$\Rightarrow t_b = \frac{\ddot{\theta}t_f - \sqrt{\ddot{\theta}^2t_f^2 - 4\ddot{\theta}(\theta_f - \theta_0)}}{2\ddot{\theta}}$$

判別式內需為正數或0，所得出 t_b

才為實數： $\ddot{\theta} \geq \frac{4(\theta_f - \theta_0)}{t_f^2}$

$$\ddot{\theta}_{min} = \frac{4(\theta_f - \theta_0)}{t_f^2}$$



Linear Function with Parabolic Blends說明-4

□ 加速度 $\ddot{\theta}$ 狀態討論

◆ If $\ddot{\theta} = \ddot{\theta}_{min}$

$t_b = \frac{t_f}{2} = t_h \Rightarrow$ 無linear線段, 兩個parabolic曲線段相連

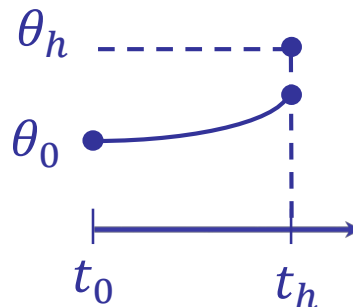
$$\text{at } t_b : \dot{\theta}(t_b) = \ddot{\theta} t_b = \frac{4(\theta_f - \theta_0)}{t_f^2} \frac{t_f}{2} = 2 \frac{\theta_f - \theta_0}{t_f}$$

速度為和「原本無規劃直接相連的速度： $\dot{\theta} = \frac{\theta_f - \theta_0}{t_f}$ 」相比為2倍

◆ If $\ddot{\theta} < \ddot{\theta}_{min}$

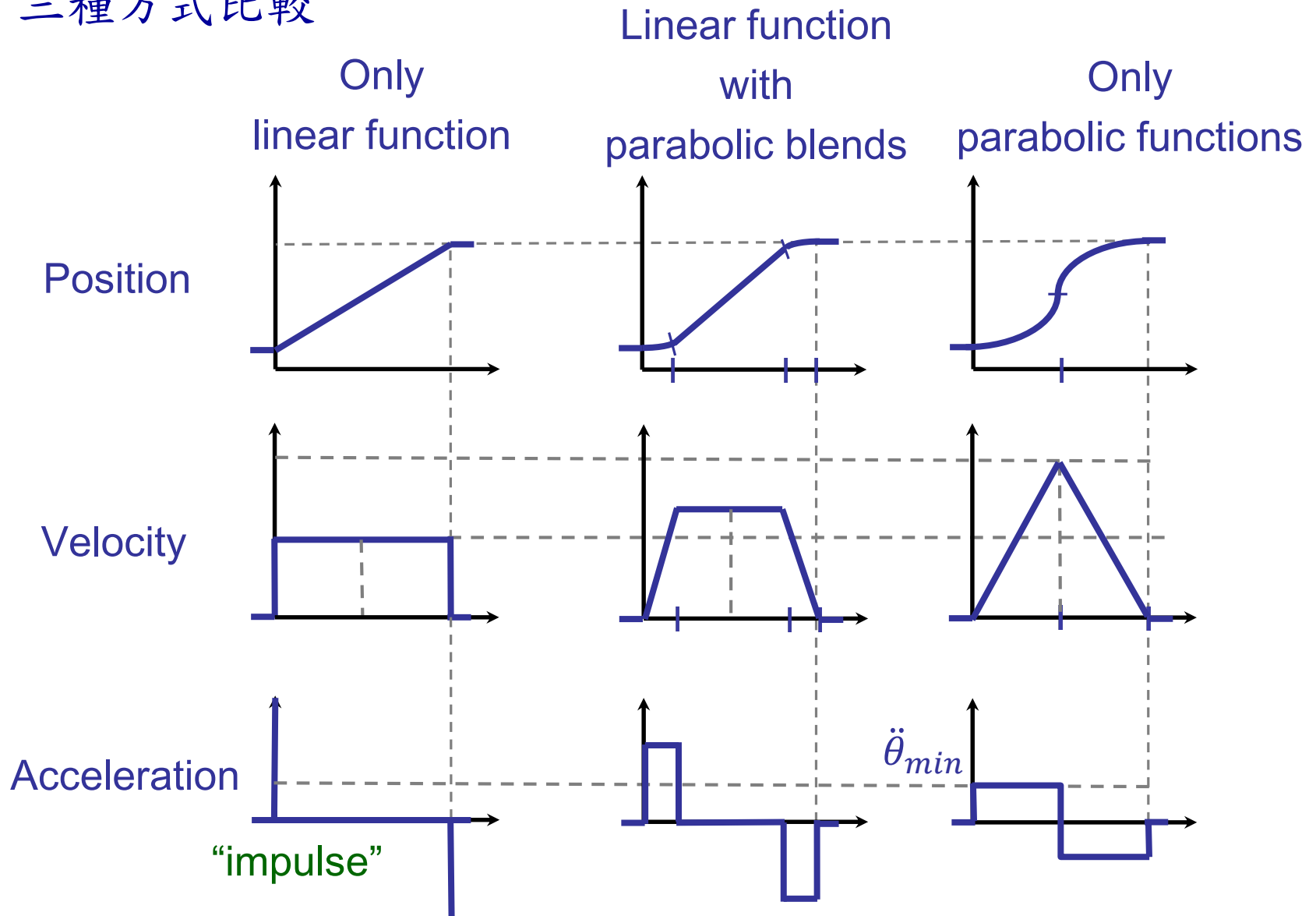
加速度不足

$$\text{at } t_b = t_h, \quad \theta < \theta_h$$



Linear Function with Parabolic Blends 說明-5

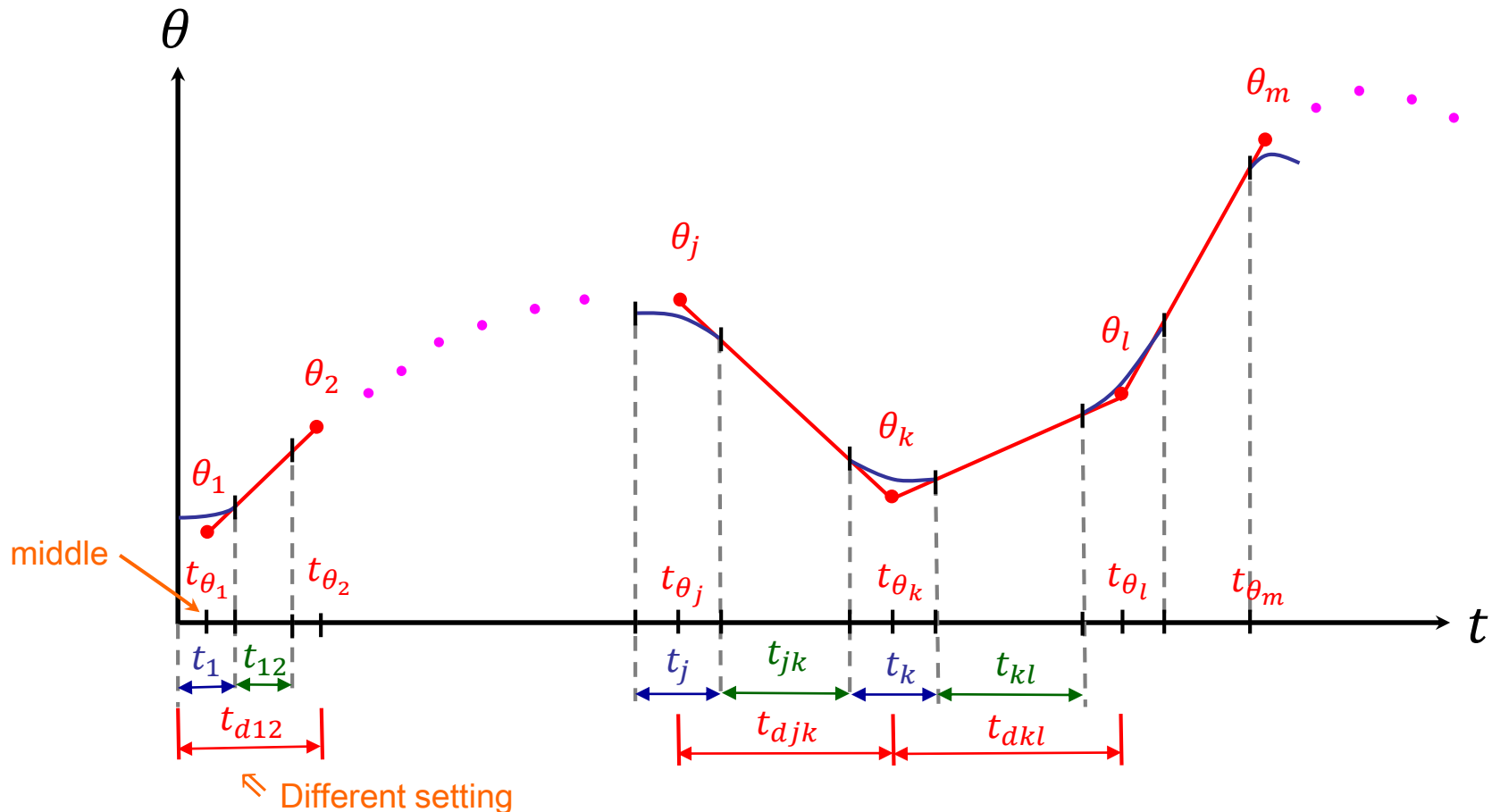
□ 三種方式比較



多段Linear Function with Parabolic Blends -1

□ 設定：A path with n via points

- ◆ 將每一個區段 $[\theta_i \theta_{i+1}]$ 各自等效到之前舉例單一linear線段 $[\theta_0 \theta_f]$ ，但與此線段前後相連接線段的速度不為0



多段Linear Function with Parabolic Blends -2

- ◆ 對任一線段 $[\theta_i \theta_{i+1}]$
 - linear (直線，一次多項式)

$$\dot{\theta}_{jk} = \frac{\theta_k - \theta_j}{t_{djk}} \quad \dot{\theta}_{kl} = \frac{\theta_l - \theta_k}{t_{dkl}}$$

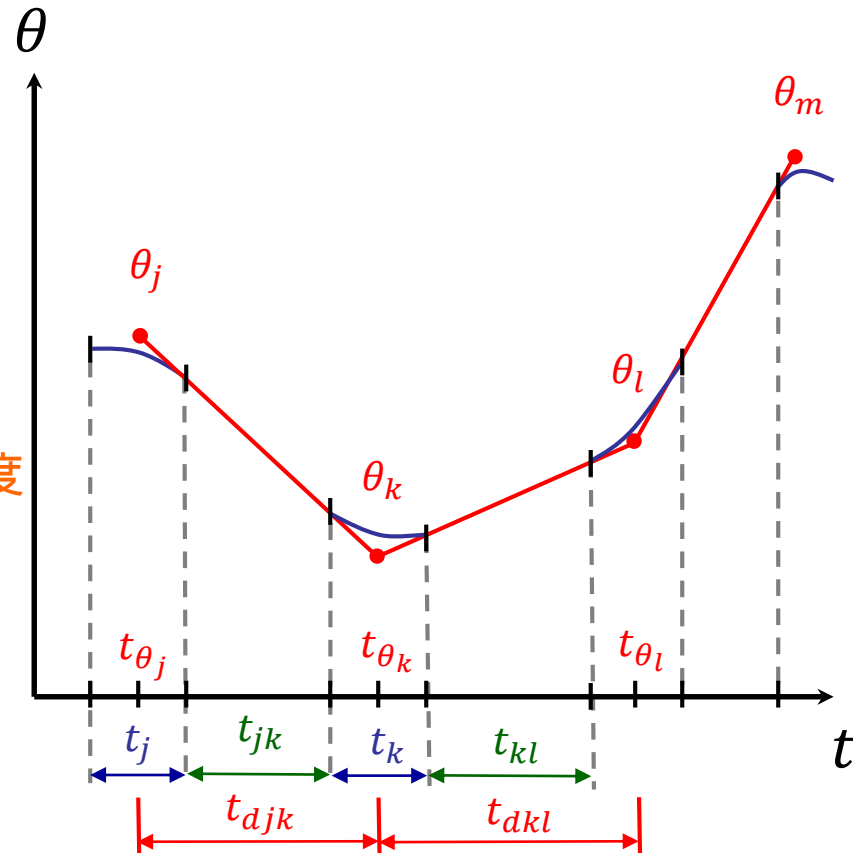
- Parabolic (二次多項式)

方法一：設定加速度解時間

$$\ddot{\theta}_k = \text{sgn}(\dot{\theta}_{kl} - \dot{\theta}_{jk}) |\ddot{\theta}_k|$$
$$t_k = \frac{\dot{\theta}_{kl} - \dot{\theta}_{jk}}{\ddot{\theta}_k} \quad \text{設定加速度}$$

方法二：設定時間解加速度

$$\ddot{\theta}_k = \frac{\dot{\theta}_{kl} - \dot{\theta}_{jk}}{t_k} \quad \text{設定時間}$$
$$t_{jk} = t_{djk} - \frac{1}{2}t_j - \frac{1}{2}t_k$$



多段Linear Function with Parabolic Blends -3

◆ 第一個線段

- θ_1 可視為整段軌跡起始點 θ_0 在時間上往後移（parabolic 曲線段所需時間 t_1 的一半），以導入 parabolic 曲線段，讓速度由起始點開始可以連續

方法一：設定加速度解時間 設定加速度

$$\ddot{\theta}_1 = \text{sgn}(\theta_2 - \theta_1) |\ddot{\theta}_1|$$

$$\dot{\theta}_{12} = \frac{\theta_2 - \theta_1}{t_{d12} - \frac{1}{2}t_1} = \ddot{\theta}_1 t_1$$

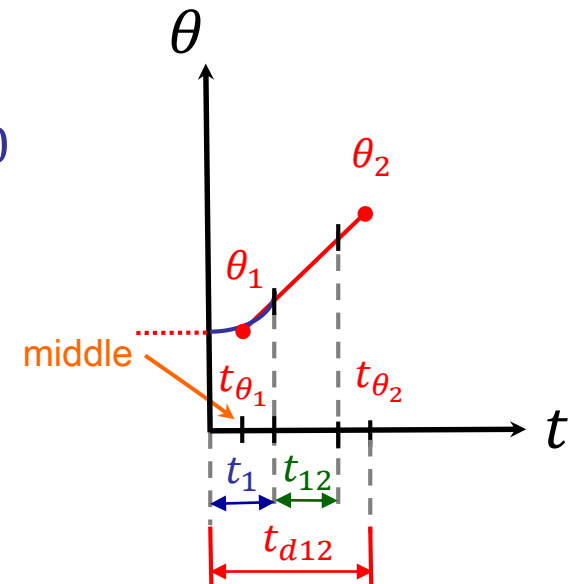
$$\frac{1}{2} \ddot{\theta}_1 t_1^2 - t_{d12} \ddot{\theta}_1 t_1 + (\theta_2 - \theta_1) = 0$$

$$t_1 = t_{d12} - \sqrt{t_{d12}^2 - \frac{2(\theta_2 - \theta_1)}{\ddot{\theta}_1}}$$

方法二：設定時間解加速度

$$\ddot{\theta}_1 = \frac{\theta_2 - \theta_1}{(t_{d12} - \frac{1}{2}t_1) t_1} \quad \text{設定時間}$$

$$t_{12} = t_{d12} - t_1 - \frac{1}{2}t_2$$



多段Linear Function with Parabolic Blends -4

◆ 最後一個線段

- θ_n 可視為整段軌跡起始點 θ_f 在時間上往前移（parabolic 曲線段所需時間 t_n 的一半），以導入 parabolic 曲線段，讓速度由起始點開始可以連續

方法一：設定加速度解時間 設定加速度

$$\ddot{\theta}_n = \text{sgn}(\theta_n - \theta_{n-1}) |\ddot{\theta}_n|$$

$$\dot{\theta}_{(n-1)n} = \frac{\theta_n - \theta_{n-1}}{t_{d(n-1)n} - \frac{1}{2}t_n} = \ddot{\theta}_n(-t_n)$$

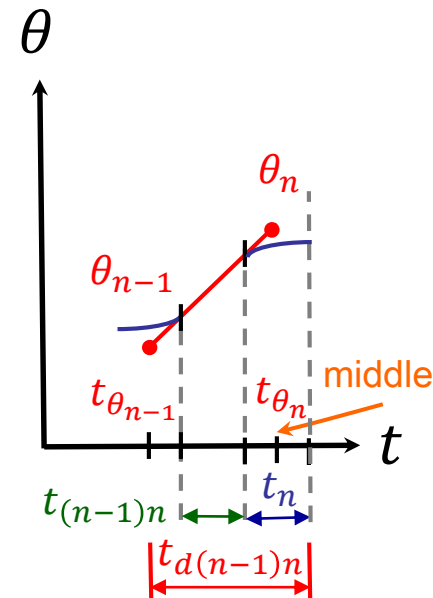
$$\frac{1}{2}\ddot{\theta}_n t_n^2 - t_{d(n-1)n}\ddot{\theta}_n t_n + (\theta_n - \theta_{n-1}) = 0$$

$$t_n = t_{d(n-1)n} - \sqrt{t_{d(n-1)n}^2 - \frac{2(\theta_n - \theta_{n-1})}{\ddot{\theta}_n}}$$

方法二：設定時間解加速度

$$\ddot{\theta}_n = \frac{\theta_n - \theta_{n-1}}{(t_{d(n-1)n} - \frac{1}{2}t_n) - t_n} \quad \text{設定時間}$$

$$t_{(n-1)n} = t_{d(n-1)n} - t_n - \frac{1}{2}t_{n-1}$$



Linear Function with Parabolic Blends 註解 -1

□ 真實系統中可達到的加速度 $\ddot{\theta}$ 取決於許多因素

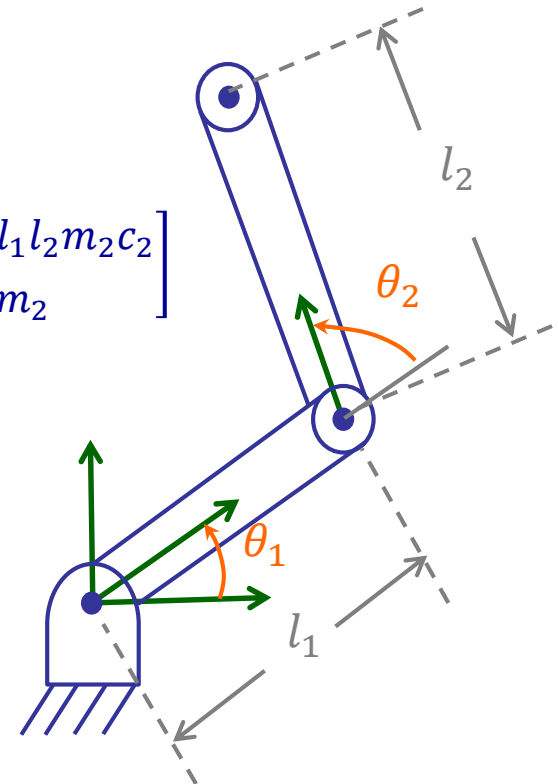
- ◆ 馬達規格
- ◆ 手臂姿態：手臂在不同姿態下，各軸所需承載（如重力）的扭力不同
- ◆ 手臂動態狀態：手臂在不同動態下，各軸需承載慣性力不同

$$\tau = M(\Theta)\ddot{\Theta} + V(\Theta, \dot{\Theta}) + G(\Theta)$$

$$M(\Theta) = \begin{bmatrix} l_2^2 m_2 + 2l_1 l_2 m_2 c_2 + l_1^2 (m_1 + m_2) & l_2^2 m_2 + l_1 l_2 m_2 c_2 \\ l_2^2 m_2 + l_1 l_2 m_2 c_2 & l_2^2 m_2 \end{bmatrix}$$

$$V(\Theta, \dot{\Theta}) = \begin{bmatrix} -m_2 l_1 l_2 s_2 \dot{\theta}_2^2 - 2m_2 l_1 l_2 s_2 \dot{\theta}_1 \dot{\theta}_2 \\ m_2 l_1 l_2 s_2 \dot{\theta}_1^2 \end{bmatrix}$$

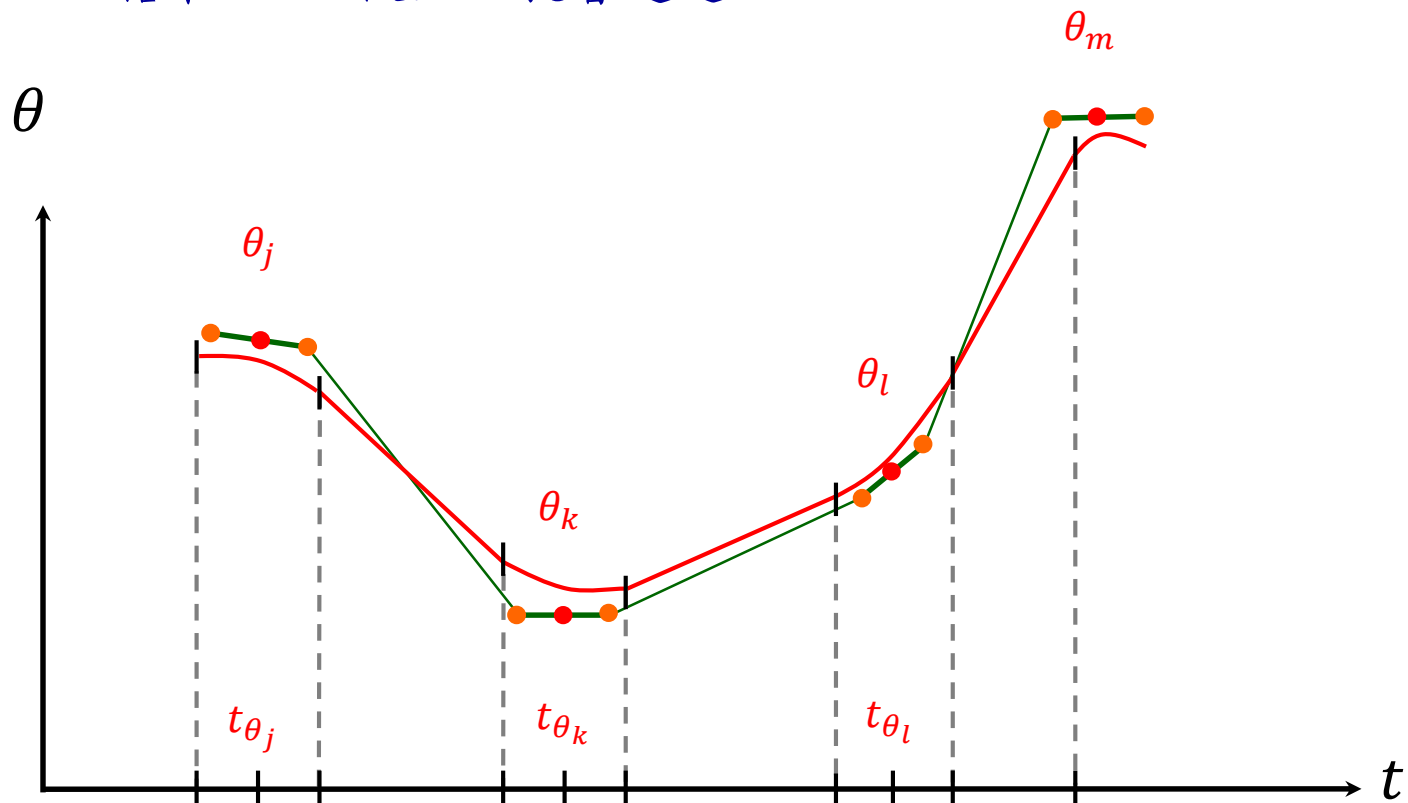
$$G(\Theta) = \begin{bmatrix} m_2 l_2 g c_{12} + (m_1 + m_2) l_1 g c_1 \\ m_2 l_2 g c_{12} \end{bmatrix}$$



Linear Function with Parabolic Blends 註解 -2

□ 規劃後軌跡並未通過via points

- ◆ 僅加速度 $\rightarrow \infty$ 的軌跡有通過via points
- ◆ 如果通過via points為必須 $\Rightarrow$ 建立pseudo via points，讓原本via points落在linear段上，就會通過



Linear Function with Parabolic Blends 註解 -3

- 若有 Cartesian space 下直線軌跡的需求，軌跡規劃需在 Cartesian space 下進行
- Programming 裡，仔細定義好某時間 t 所屬的線段或曲線段

- ◆ Linear: $t \in [t_{\theta_j} + \frac{1}{2}t_j \quad t_{\theta_k} - \frac{1}{2}t_k]$

$$\theta(t) = \theta_j + \dot{\theta}_{jk}\Delta t = \theta_j + \dot{\theta}_{jk}(t - t_{\theta_j})$$

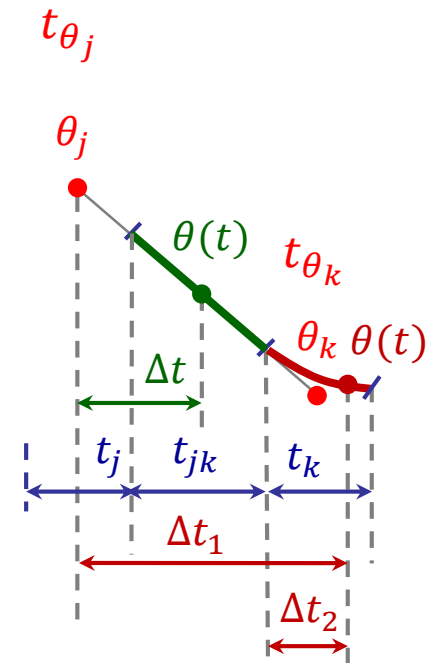
$$\dot{\theta}(t) = \dot{\theta}_{jk} \quad \ddot{\theta} = 0$$

- ◆ Parabolic: $t \in [t_{\theta_k} - \frac{1}{2}t_k \quad t_{\theta_k} + \frac{1}{2}t_k]$

$$\theta(t) = \theta_j + \dot{\theta}_{jk}\Delta t_1 + \frac{1}{2}\ddot{\theta}_k \Delta t_2^2$$

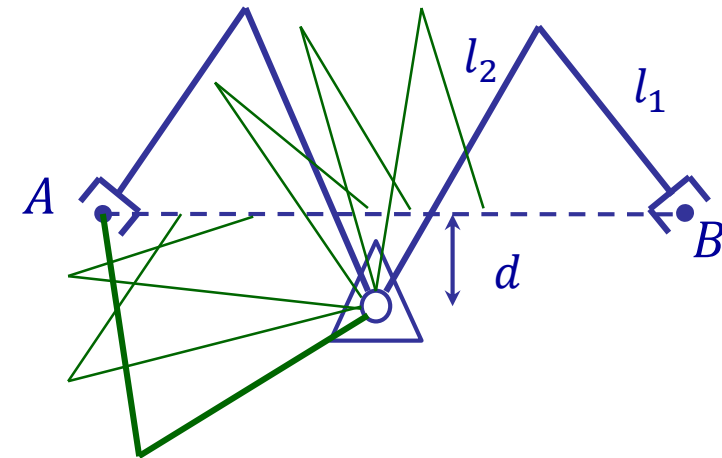
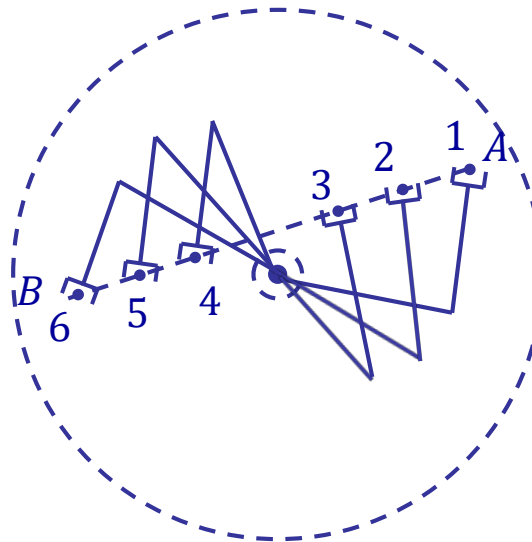
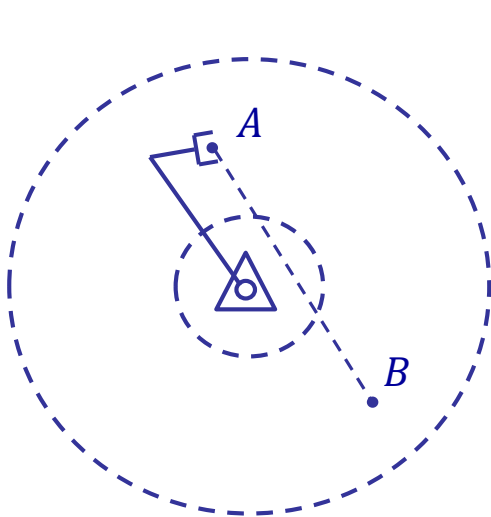
$$= \theta_j + \dot{\theta}_{jk} \left(t - t_{\theta_j} \right) + \frac{1}{2}\ddot{\theta}_k \left(t - t_{\theta_k} + \frac{1}{2}t_k \right)^2$$

$$\dot{\theta}(t) = \dot{\theta}_{jk} + \ddot{\theta}_k \left(t - t_{\theta_k} + \frac{1}{2}t_k \right) \quad \ddot{\theta}(t) = \ddot{\theta}_k$$



Cartesian Space 下軌跡幾何限制

- 中間某些區段無法到達 (i.e., 在work space之外)
- 軌跡需要高加減速 (i.e., 接近singular configuration)
- 針對特定起使和終點姿態，無法產生連續軌跡
 - ◆ 如下圖，除非 $l_1 - l_2 = d$



Example: Revisit the RRR Manipulator -1

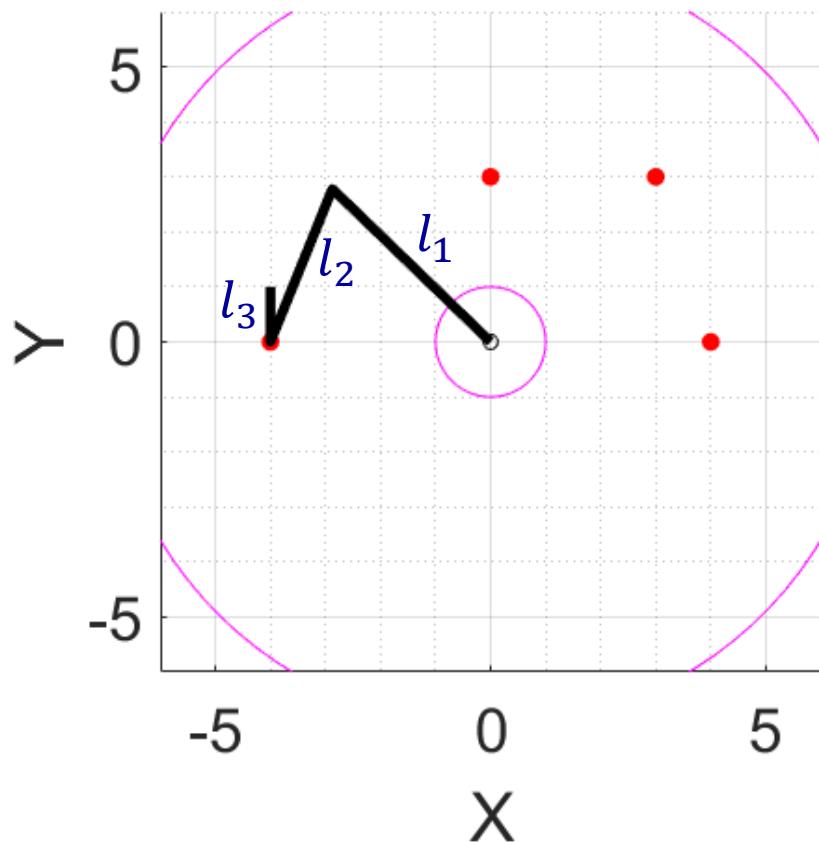
- 平面RRR手臂長度： $l_1 = 4$, $l_2 = 3$, and $l_3 = 1$

下表定義 initial, via, via, 和 final points的位置

- 方法：以linear function with parabolic blends在Cartesian-space下規劃軌跡

i	t_i	x_i	y_i	θ_i
0	0	-4	0	90
1	2	0	3	45
2	4	3	3	30
3	7	4	0	0

(X,Y) 定義在第二桿件的末端
 θ 為第三桿件對X座標軸的夾角



Example: Revisit the RRR Manipulator -2

1. 求出各DOF (X, Y, θ) 在每段的速度及加速度
(每段Parabolic function區間長0.5秒)

i	t _i	x _i	y _i	θ_i
0	0	-4	0	90
1	2	0	3	45
2	4	3	3	30
3	7	4	0	0



	X	Y	$\theta(\text{deg/s})$
V_0	0	0	0
V_1	2.29	1.71	-25.71
V_2	1.5	0	-7.5
V_3	0.36	-1.09	-10.9
V_f	0	0	0



	X	Y	$\theta(\text{deg/s}^2)$
a_0	4.57	3.43	-51.4
a_1	-1.57	-3.43	36.4
a_2	-2.27	-2.18	-6.82
a_f	-0.72	2.18	21.8

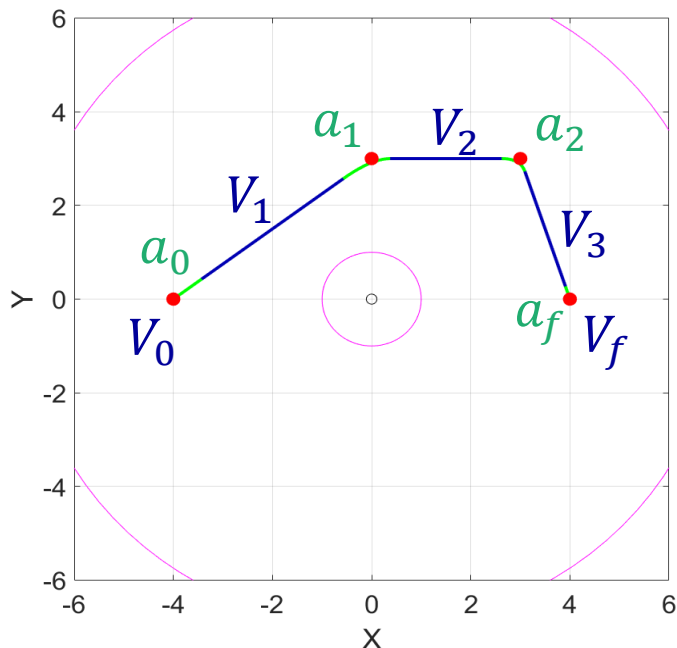
中間線段：

$$V_2 = \frac{DOF_2 - DOF_1}{4 - 2}$$

頭尾線段：

$$V_1 = \frac{DOF_1 - DOF_0}{2 - 0 - \frac{0.5}{2}}$$

$$V_3 = \frac{DOF_3 - DOF_2}{7 - 4 - \frac{0.5}{2}}$$



$$a_0 = \frac{V_1 - V_0}{0.5}$$

$$a_1 = \frac{V_2 - V_1}{0.5}$$

$$a_2 = \frac{V_3 - V_2}{0.5}$$

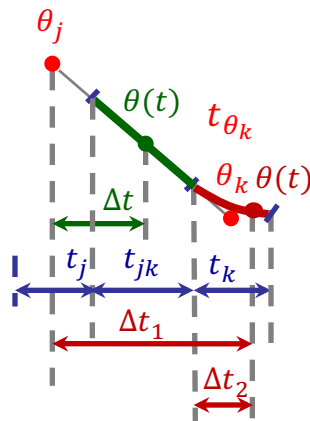
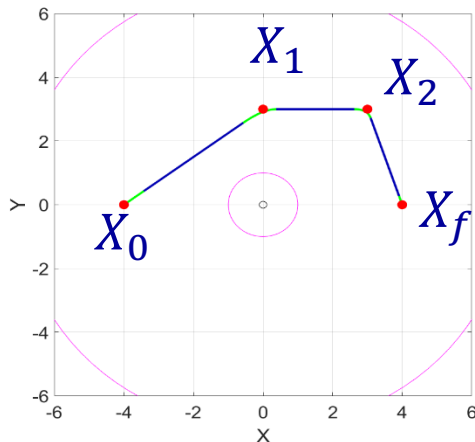
$$a_3 = \frac{V_f - V_3}{0.5}$$

Example: Revisit the RRR Manipulator -3

2. 建立各DOF (X, Y, θ) 在每段的equation (Linear/Parabolic 共7段)

以X為例:

$X_{eq1}(t) = X_0 + V_0 \Delta t + \frac{1}{2} a_0 \Delta t^2$	$= -4 + 0(t - 0) + \frac{1}{2} 4.57(t - 0)^2$	$t \in [0, 0.5]$
$X_{eq2}(t) = X_0 + V_1 \Delta t$	$= -4 + 2.29(t - 0.25)$	$t \in [0.5, 1.75]$
$X_{eq3}(t) = X_0 + V_1 \Delta t_1 + \frac{1}{2} a_1 \Delta t_2^2$	$= -4 + 2.29(t - 0.25) + \frac{1}{2} (-1.57)(t - 1.75)^2$	$t \in [1.75, 2.25]$
$X_{eq4}(t) = X_1 + V_2 \Delta t$	$= 0 + 1.5(t - 2)$	$t \in [2.25, 3.75]$
$X_{eq5}(t) = X_1 + V_2 \Delta t_1 + \frac{1}{2} a_2 \Delta t_2^2$	$= 0 + 1.5(t - 2) + \frac{1}{2} (-2.27)(t - 3.75)^2$	$t \in [3.75, 4.25]$
$X_{eq6}(t) = X_2 + V_3 \Delta t$	$= 3 + 0.36(t - 4)$	$t \in [4.25, 6.5]$
$X_{eq7}(t) = X_2 + V_3 \Delta t_1 + \frac{1}{2} a_3 \Delta t_2^2$	$= 3 + 0.36(t - 4) + \frac{1}{2} (-0.73)(t - 6.5)^2$	$t \in [6.5, 7]$



Recall:

Linear:

$$\theta(t) = \theta_j + \dot{\theta}_{jk} \Delta t = \theta_j + \dot{\theta}_{jk} (t - t_{\theta_j})$$

Parabolic:

$$\begin{aligned} \theta(t) &= \theta_j + \dot{\theta}_{jk} \Delta t_1 + \frac{1}{2} \ddot{\theta}_k \Delta t_2^2 \\ &= \theta_j + \dot{\theta}_{jk} (t - t_{\theta_j}) + \frac{1}{2} \ddot{\theta}_k (t - t_{\theta_k} + \frac{1}{2} t_k)^2 \end{aligned}$$

Example: Revisit the RRR Manipulator -4

3. 繪出各DOFs (X , Y , θ) 之規劃軌跡

4. 以IK計算3軸轉角，將手臂參數帶入FK，畫出手臂的空間運動軌跡，以確認軌跡規劃的正確性

